ATAR course **Year 12 Physics**

Student Name: ____________________

Practice Exam 1 (Units 3 & 4)

Time allowed

Reading time before commencing work: 10 minutes

Working time: three hours

Section	Number of questions	Number of questions to be answered	Suggested working time (minutes)	Number of marks	Percentage of exam
Section one: Short answers	13	13	50	57	30%
Section two: Problem solving	6	6	90	82	50%
Section three: Comprehension	2	2	40	37	20%
Totals	21	**21**	**180**	**176**	**100%**

Notes to students

- Write your name in the space above.
- A Formulae and Data Booklet has been provided.
- The following items are approved for use in the examinations:

  Standard items: pens (blue/black preferred), pencils (including coloured), sharpener, correction fluid/tape, eraser, ruler, highlighters.

  Special items: non-programmable calculators approved for use in the WACE examinations, drawing templates, drawing compass and a protractor
- You must be careful to confine your responses to the specific questions asked and to follow any instructions that are specific to a particular question.

Disclaimer:

This is a practice examination. It represents Pearson Australia's view only of what would be useful preparation for the externally assessed examination.

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Section one: Short answer 30% (57 marks)

This section has 13 questions. Answer **all** questions. Write your answer in the spaces provided.

When calculating numerical answers, show your working or reasoning clearly. Give final answers to **three** significant figures and include appropriate units where applicable.

When estimating numerical answers, show your working or reasoning clearly. Give final answers to **two** significant figures and include appropriate units where applicable.

Spare pages are included at the end of this booklet. The can be used for planning your responses and/or as additional space if required to answer a question.

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- Continuing an answer: If you need to use the space to continue an answer, indicate in the original answer space where the answer is to be continued (i.e. give the page number). Fill in the number of the question you are continuing to answer at the top of the page.

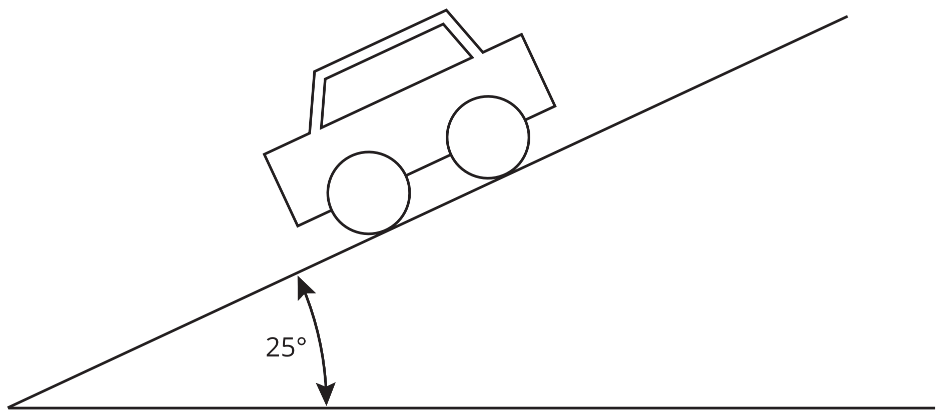
Suggested working time: 60 minutes

Question 1 **(4 marks)**

Gerald and Omar are standing on a first-floor balcony. Gerald throws a ball horizontally and at the same instant Omar drops an identical ball. Sally, who is standing below, observes the balls hit the ground at the same time. Explain this observation. Identify the relevant mathematical relationship in your explanation.

Question 2 **(5 marks)**

A car of mass 1540 kg is stationary on a hill that has a slope of 25° as represented below.



Sketch and label the forces acting on the stationary car on the diagram given, and calculate the force needed to stop the car from moving.

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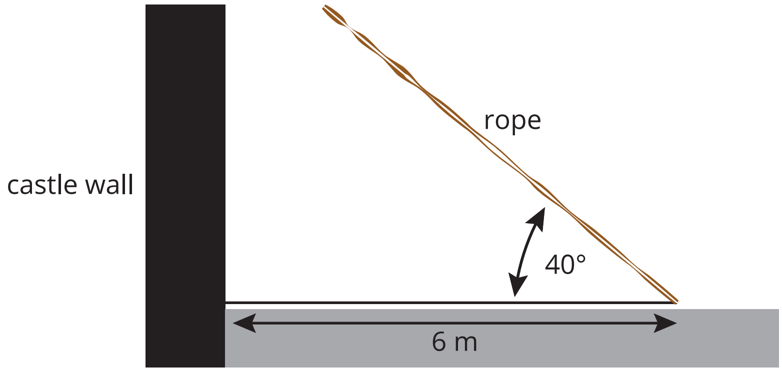
Question 3 **(4 marks)**

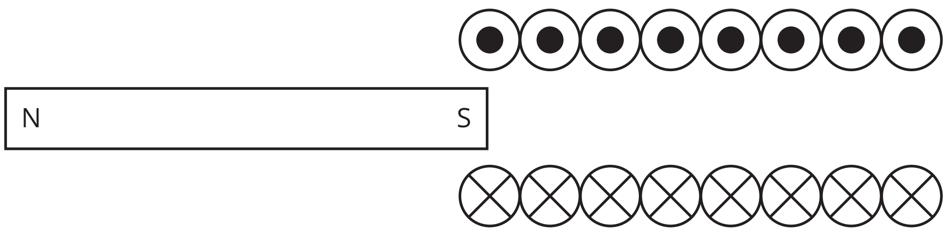
An 8 m castle drawbridge of uniform mass 3200 kg is raised by a rope attached at an angle of 40° and 6 m from its end as shown below. Determine the tension (force) in the rope needed to begin moving the drawbridge from its horizontal position.



Question 4 **(5 marks)**

a Complete the diagram below by drawing at least **five** field lines around the conductors and state whether the wires are attracted towards or repelled from each other. (3 marks)

b Complete the diagram below by showing the direction of movement of the magnet that generates the current in the solenoid represented below. (1 mark)



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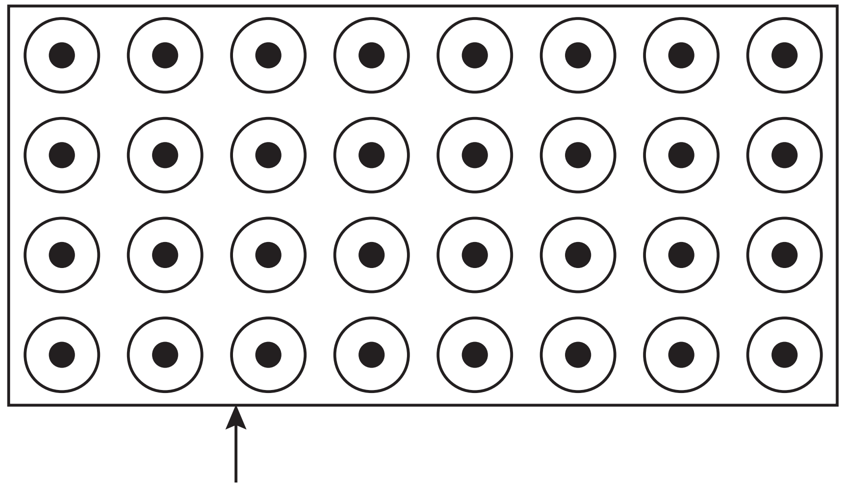
c Complete the diagram by drawing the path a positively charged particle will take on entering the magnetic field shown below. (1 mark)



Question 5 **(4 marks)**

What is the wavelength of the radiation emitted from 50 Hz electric power lines? What is the energy of a photon of this wavelength?

Question 6 **(4 marks)**

Einstein's special theory of relativity is based on the two principles that:

- the speed of light in a vacuum is an absolute constant, and
- all inertial frames of reference are the same.

Identify the apparent contradiction of accepting these two principles and explain how Einstein's special theory of relativity resolves this apparent contradiction.

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Question 7 **(5 marks)**

The table below gives leptons and their antiparticles.

Lepton	Lepton symbol	Antilepton symbol
electron	e^-	e^+
electron neutrino	ν_e	$\bar{\nu}_e$
muon	μ^-	μ^+
muon neutrino	ν_μ	$\bar{\nu}_\mu$
tau	τ^-	τ^+
tau neutrino	ν_τ	$\bar{\nu}_\tau$

All leptons have a lepton number of positive one (i.e. $L = +1$) and antileptons have a lepton number of negative one (i.e. $L = -1$).

Use the above information to answer the following questions.

a Determine whether the following decay for a muon is possible. Explain your answer. (3 marks)

$$\mu^- \rightarrow e^- + \nu_e$$

b When a muon decays to **three** other leptons and/or antileptons, what is the only possible combination? Give your reasoning. (2 marks)

Note: You are not required to identify a specific set of leptons and/or antileptons as decay products but rather a general statement about the number of leptons and/or antileptons permissible when a muon decays to a total of three products.

Question 8 **(5 marks)**

The electric trains in Perth use some of their electric motors as generators when breaking. Briefly describe how an electric generator produces a current and explain why turning the current off to the motor and allowing it to operate as a generator helps slow the train.

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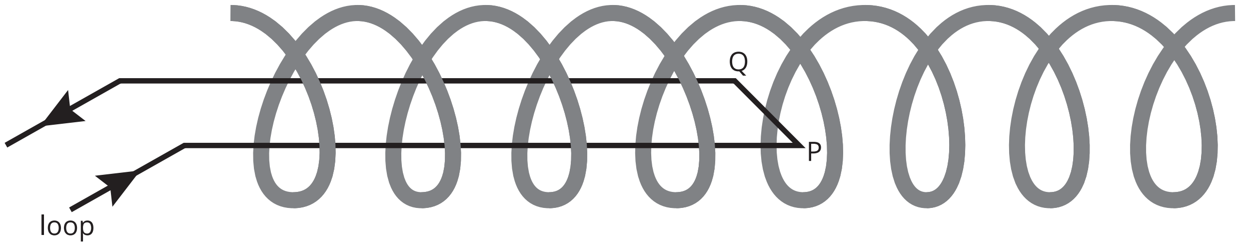
Question 9 **(4 marks)**

A conductor loop is positioned inside a solenoid with its long axis parallel to the axis of the solenoid, as shown in the diagram below. The short side PQ is at right angles to the axis and is 25 mm long. The current in the loop is 5.0 A, while the current in the solenoid produces a magnetic field inside the solenoid with flux density of 40.0 mT to the right.



a What is the direction of the magnetic force on the side PQ? (1 mark)

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b What is the magnitude of this force on the side P? (3 marks)

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Question 10 **(5 marks)**

The Australian synchrotron is a light source synchrotron. It accelerates electrons to high speed and then converts this energy to a range of different wavelength electromagnetic radiation (light) to conduct experiments.

a The synchrotron light is produced when the electrons are forced to change direction by an external magnetic field. Explain how a magnetic field causes the electrons to change direction. (2 marks)

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b If an electron accelerated to have 2.5 GeV of energy has 0.01% of its energy converted to light (electromagnetic radiation), what is the wavelength of the light? (3 marks)

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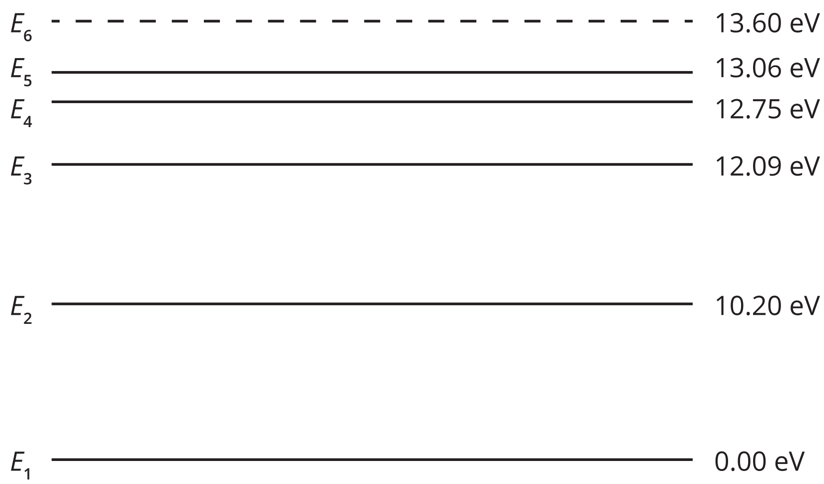
Question 11 **(4 marks)**

What is the wavelength of an electron accelerated to 0.85 the speed of light (i.e. 0.85c)?

Question 12 **(4 marks)**

The energy level diagram for hydrogen shown below can be used to answer the following question.



A satellite equipped with an instrument to measure radiation emitted in the ultraviolet region of the electromagnetic spectrum records a wavelength of 9.76×10^{-8} m. Between which two energy levels must the hydrogen atom's electron transition to produce this line?

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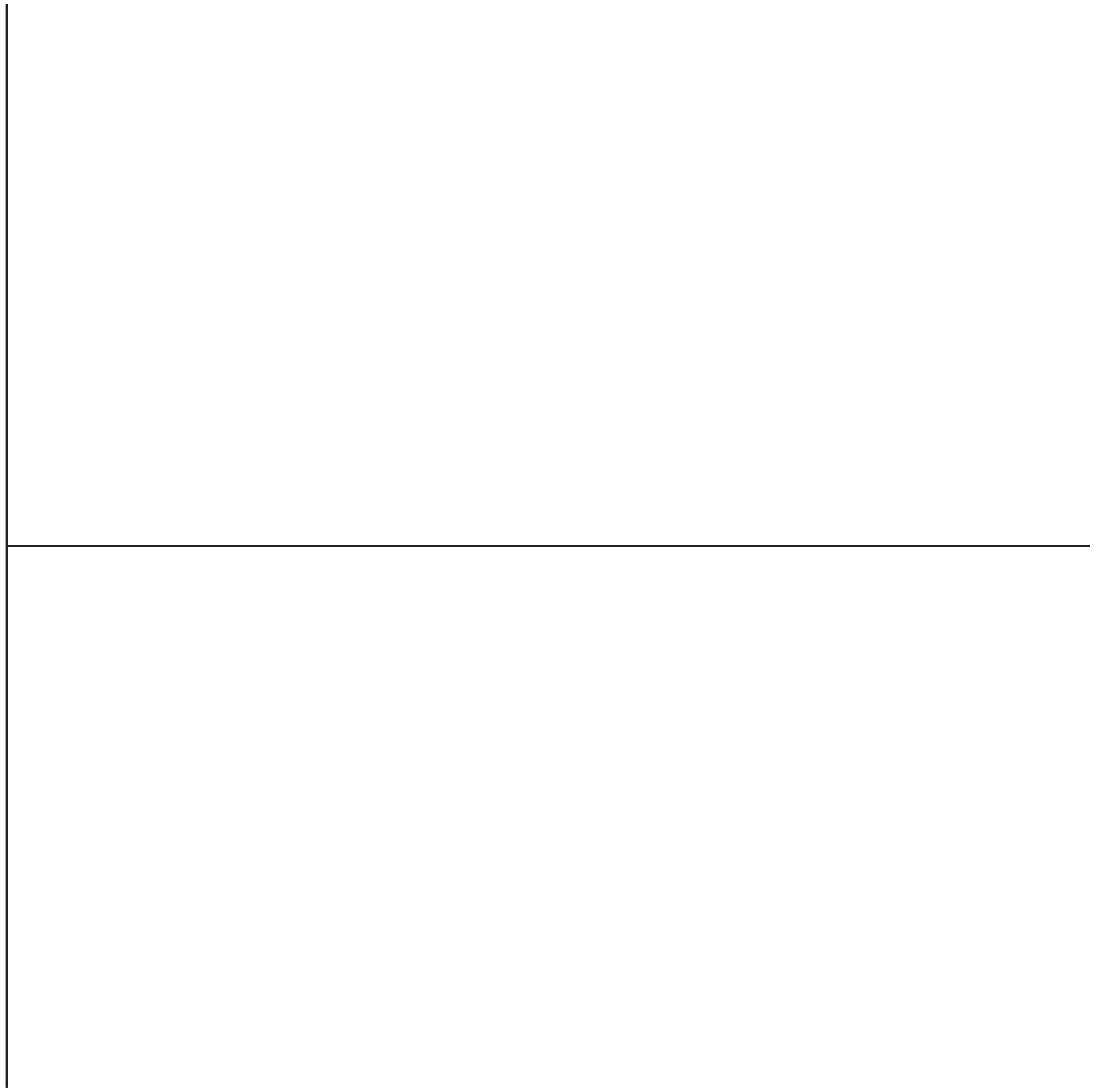
Question 13 **(4 marks)**

On the axes below, draw a graph to show the EMF for an AC generator with a frequency of 25 Hz and root mean square voltage of 85 V.



Section two: **Problem-solving** 50% (82 marks)

This section has 6 questions. Answer **all** questions. Write your answers in the spaces provided.

When calculating numerical answers, show your working or reasoning clearly. Give final answers to **three** significant figures and include appropriate units where applicable.

When estimating numerical answers, show your working or reasoning clearly. Give final answers to a maximum of **two** significant figures and include appropriate units where applicable.

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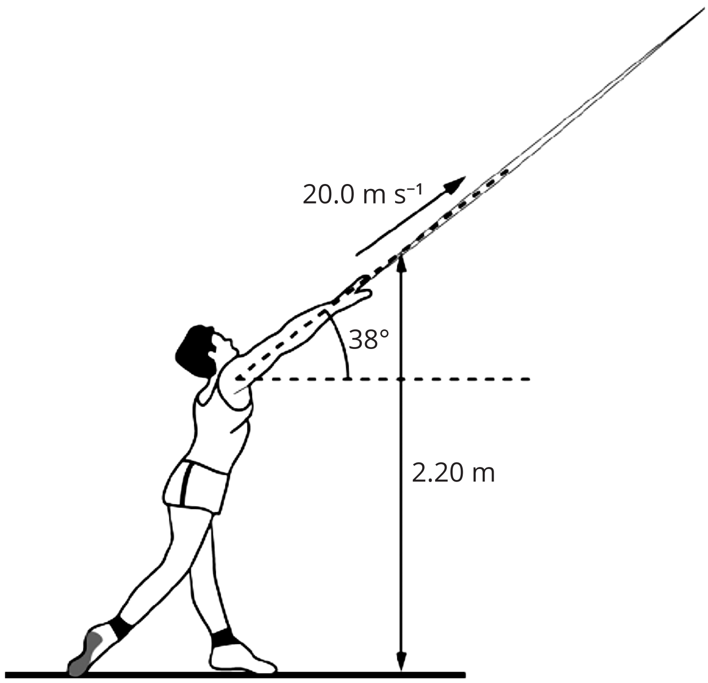
Suggested working time: 90 minutes.

Question 14 **(13 marks)**

During a school athletics competition, an athlete threw a javelin as shown below.



At the moment it left the athlete's hand, the 0.85 kg javelin was at a height of 2.2 m above the ground and travelling at 20 m s^{-1} at an angle of elevation of 38°.

a What was the maximum height above the ground reached by the javelin? Ignore the effects of air resistance. (3 marks)

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b How far horizontally from its point of release did the javelin land? Ignore the effects of air resistance. (5 marks)

c How much kinetic energy does the javelin have when it strikes the ground at the end of its flight? (5 marks)

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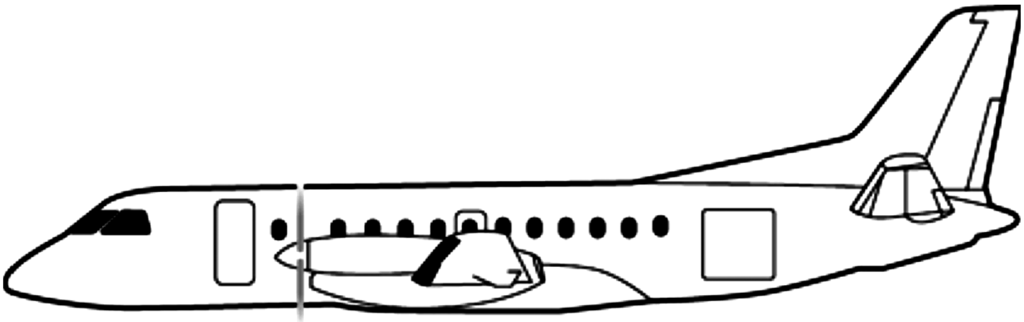
Question 15 **(14 marks)**

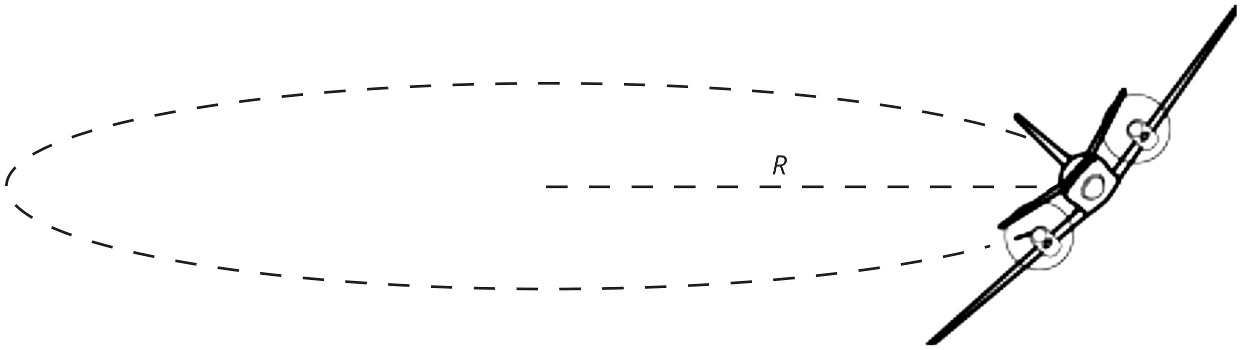
A pilot of mass 75 kg was flying in an aeroplane as shown below.



The combined mass of the pilot and aeroplane was 5000 kg.

Initially, the pilot flew in a horizontal circle as shown by the diagrams in parts a and b of this question.

a While flying in a horizontal circle, as represented below, the aeroplane flew at a *constant* speed of 120 m s^{-1} and it took 20 seconds for it to complete one full circle. What was the radius, R, of the horizontal circle that the aeroplane flew around? (2 marks)



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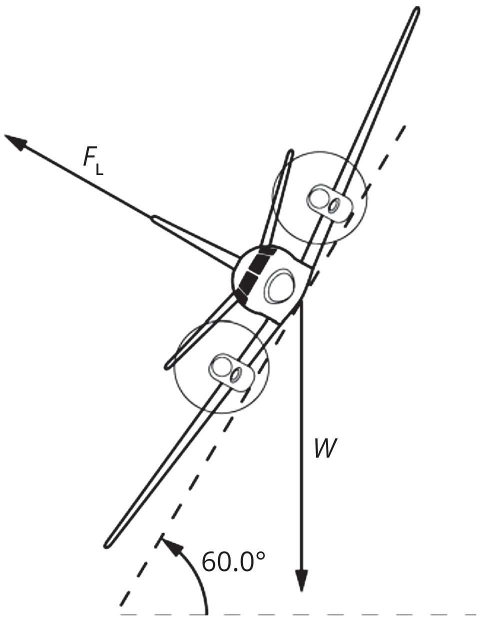
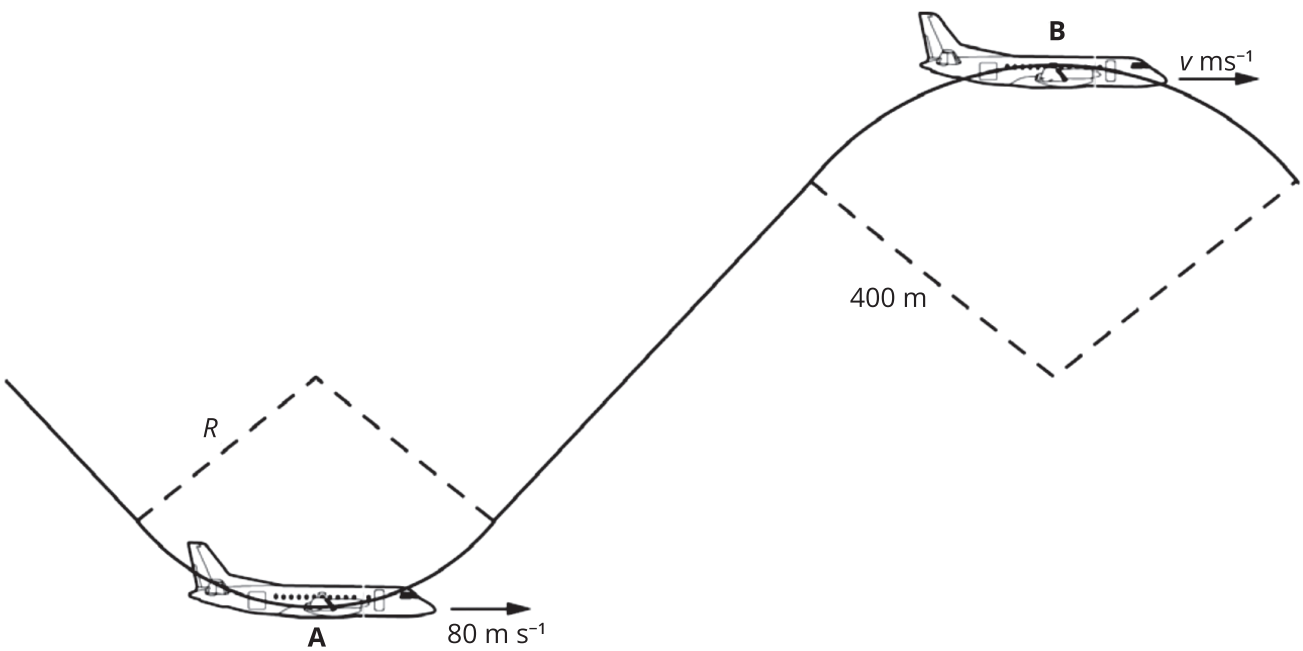
b The figure below shows the only two unbalanced forces that were acting on the aeroplane during the manoeuvre to fly the horizontal circle shown in part a of this question. Calculate the magnitude of the lift force, F_L. (3 marks)



A short time later, the pilot performed the aerobatics as represented in the figure below. This time the manoeuvres involved flying along the arcs of two vertical circles shown at A and B.

A shows the aeroplane at the bottom of an arc of radius R flying at a constant horizontal speed of 80 m s^{-1}.

B shows the aeroplane at the top of an arc of radius 400 metres flying at an unknown constant horizontal speed of v m s^{-1}.



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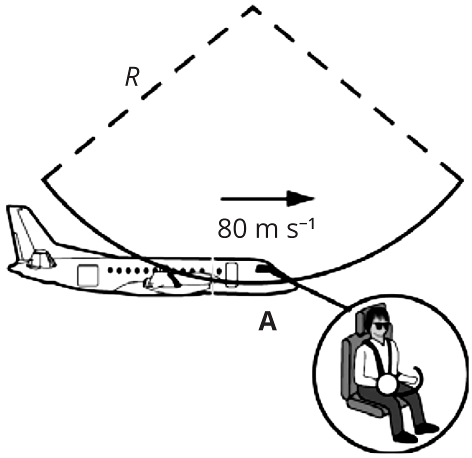
c At the point A as shown below, the pilot's apparent weight is 3000 N. Draw and label the direction and magnitude of the forces acting on the pilot at this instant on the diagram provided. (2 marks)



d If the aeroplane was flying at a constant speed of 80 m s^{-1} at point A, what would be the radius, *R*, of the arc flown? (4 marks)

e At point B the pilot experiences 'apparent weightlessness' or freefall. At what speed would the aeroplane need to be flying at point B in order for the pilot to experience 'apparent weightlessness'? (3 marks)

Question 16 **(12 marks)**

A weather satellite is placed in orbit 600 km above the Earth's surface.

a What orbital velocity must this satellite have? (3 marks)

b Calculate the period of orbit of the satellite. (2 marks)

c What acceleration due to the Earth's gravitational field does the satellite experience? (3 marks)

d Why do objects in the satellite experience weightlessness? (2 marks)

e Orbiting satellites are fitted with small propulsion devices to assist in stabilising their orbit. Without these devices, they are likely to slow. Explain why the rockets may slow and what would happen if they did. (2 marks)

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Question 17 **(14 marks)**

In an experiment to determine the threshold frequency and work function for a particular metal, a researcher measured the kinetic energy of the electrons ejected from the metal's surface at a range of frequencies. The results are tabulated below.

Frequency ($\times 10^{15}$) (Hz)	Kinetic energy of electron ($\times 10^{-19}$) (J)
2.50	9.055
5.00	25.63
7.50	42.20
1.00	58.78
1.25	75.36

a Plot a graph of frequency against kinetic energy on the axis provided below. (5 marks)

b Use data from the graph to determine the threshold frequency for the metal. (2 marks)

c Use data from the graph to determine the work function for the metal. (2 marks)

d Determine the slope of the line. (2 marks)

The researcher carries out the same experiment to determine the threshold frequency and work function for a second metal, and graphs the frequency against kinetic energy.

e Predict the slope of the line for this second metal. (1 mark)

f Explain what the effect studied in this experiment provides evidence for in respect of electromagnetic radiation. (2 marks)

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Question 18 **(13 marks)**

A space probe is sent to make observations of a planet orbiting a star in a neighbouring galaxy. The planet is 24.6 light-years from Earth and the probe is sent from Earth with a speed $0.82c$. The probe carries an atomic clock that was synchronised with a similar clock on Earth at the start of the journey.

a How long will the journey take as measured by the clock on Earth? (2 marks)

b How long will the journey take as measured by the clock on the probe? (2 marks)

c From the space probe's frame of reference, what is the distance travelled to reach the planet? (2 marks)

d From the space probe's frame of reference, at what velocity is it travelling? (2 marks)

e On arrival, the probe launches a smaller probe towards the star around which the planet orbits at a speed of $0.650c$. What is the speed of this smaller probe from the Earth's frame of reference? (2 marks)

f On arrival at the planet, the space probe sends signals of its measurements back to Earth using various parts of the electromagnetic spectrum.

i At what velocity will the signals travel from the probe's frame of reference? (1 mark)

ii At what velocity will the signals travel from the Earth's frame of reference? (1 mark)

iii Explain your answers to (i) and (ii). (1 mark)

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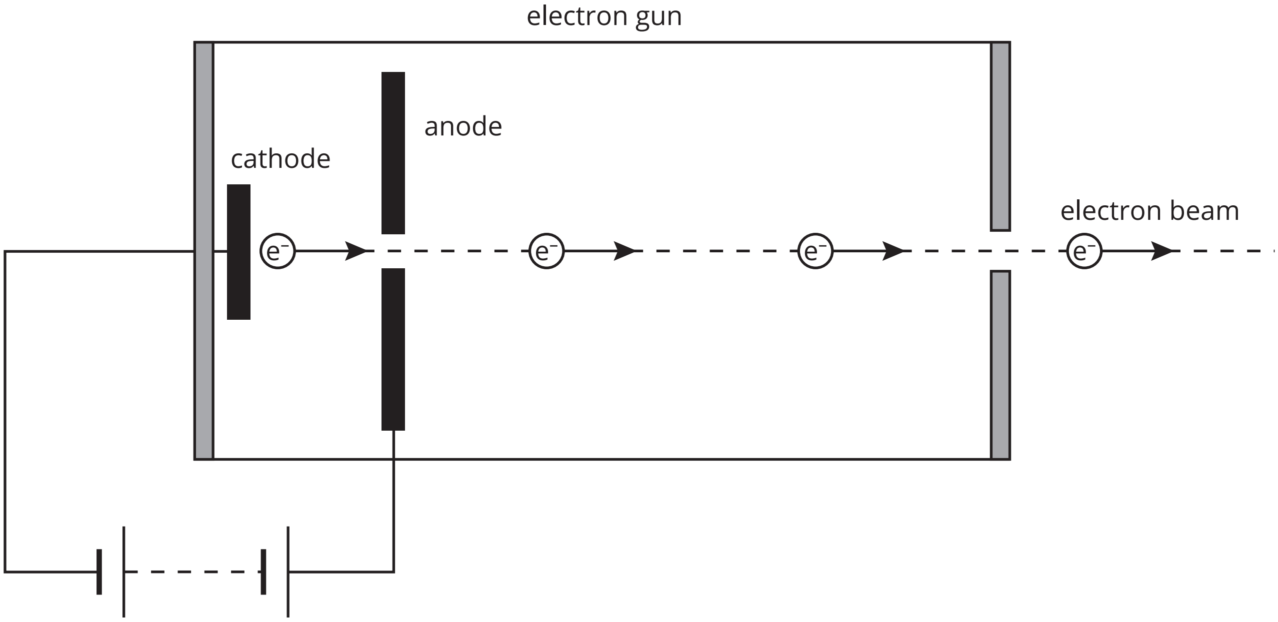
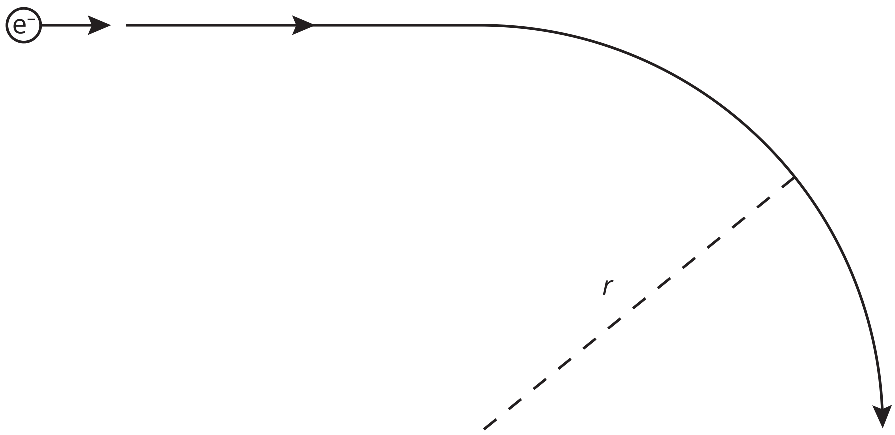
Question 19 **(16 marks)**

An electron gun is used to inject electrons into a synchrotron. The electron gun releases electrons from its cathode, which are then accelerated across a potential difference of 16 kV. The distance between the cathode and anode is 25 cm. Below is shown a schematic diagram of an electron gun.



a With what speed will the electrons exit the electron gun? (4 marks)

The electron gun's accelerating voltage was adjusted so that the exiting electrons were travelling at a speed of 1.5×10^5 m s^{-1}. The path taken by the electrons as they entered a magnetic field of strength 20 mT is shown below.



b State the direction of the magnetic field that causes the electrons to follow the path shown above. (1 mark)

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c What is the magnitude of the force acting on the electrons as they travel through the magnetic field? (2 marks)

d What is the radius of the arc of the circle that the electrons follow while they are in the magnetic field? (2 marks)

An experiment is carried out in the synchrotron where an electron collides with a positron (the antiparticle of an electron).

e The product of a low-energy collision is two photons. Draw a Feynman diagram to represent this collision. Label the diagram. (3 marks)

f To produce particles heavier than themselves, an electron and a positron need to collide at high kinetic energies. When they collide at low energy, they produce only two (or more) photons. Explain why high energies are needed for production of particles heavier than themselves. (2 marks)

g Another consequence of high-energy collisions is greater resolution to study finer structures. Explain how high-energy collisions provide higher resolution. (2 marks)

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Section three: **Comprehension** 20% (37 marks)

This section has 2 questions. Answer **both** questions. Write your answers in the spaces provided.

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Suggested working time: 40 minutes.

Question 20 **(20 marks)**

'X' for unknown

While working on cathode rays (high-energy electron beams) and their effects in 1895, Wilhelm Roentgen observed that when electrons accelerated to a high speed in a vacuum tube struck the wall of the glass tube, fluorescent minerals some distance away would glow. It was known electrons travelled only a few centimetres in air so Roentgen concluded that these effects were due to a new type of radiation he called 'X-radiation' or X-rays. Because these X-rays were not detectable by electric or magnetic fields, it was concluded they were not charged particles.

About a month after Roentgen's discovery, J. J. Thompson observed that if the X-rays were passed through a gas, the gas became electrically conducting.

It was suggested the X-rays were a form of invisible light but no refraction, interference or diffraction patterns were observed when the rays were used with ordinary optical instruments. It was noted that the available diffraction gratings at that time had a typical width of the order of 10^{-6} m. In approximately 1912, it was suggested that to test if X-rays were a wave they could be directed onto crystals, as it was estimated that atoms in a crystal were separated by about 10^{-10} m so the crystal could act as a diffraction grating. Scattering of X-rays from crystals did show diffraction patterns suggesting the wave-like nature of X-rays.

It was also observed that X-rays are produced when high-speed electrons strike a metal surface. But it was also shown that electrons are ejected from metals that are exposed to X-rays. The specific frequency of X-rays that cause the ejection of an electron is unique to each metal and more electrons are ejected with higher intensity X-rays of the required frequency.

a Explain why mineral exposed to this new radiation would fluoresce. (3 marks)

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b What **two** conclusions, other than the one given in the information, can be made about X-radiation based on the nature of the interaction of the X-radiation with electric and magnetic fields? (2 marks)

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c Why can it be concluded that the X-radiation is not a charged particle? Explain the physics principle on which this conclusion is based. (2 marks)

d Explain J. J. Thompson's observation. (3 marks)

e Why were no interference or diffraction patterns observed with the typical gratings available at the time of the discovery of the X-radiation? Estimate the wavelength of X-radiation based on the information provided. Identify the evidence in the supplied information that supports your estimate. (3 marks)

f What can be concluded about the nature of X-rays from the observation of the effect of exposing metals to X-rays? What name is given to this effect? (2 marks)

g Explain why high-energy X-rays have a smaller wavelength. Support your answer with the appropriate mathematical relationship. (2 marks)

h Tungsten is often used in X-ray tubes for making X-rays for medical purposes. What is the minimum energy an electron striking a tungsten plate needs to generate an X-ray with a wavelength of 0.0184 nm? (3 marks)

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Question 21 **(17 marks)**

What mass is and how we measure it: the electronic kilogram

Many physicists are grappling with the question of 'What is mass?' In Newtonian physics, mass is the amount of matter in an object and the concept of mass was established by Newton with his recognition of the following relationship.

$$F = ma$$

With the development of quantum physics and relativity, the Newtonian concept of mass was challenged. Einstein's now famous equation relates the rest mass of a particle to its total energy. As our understanding of the quantisation of matter and relativity developed, it was realised that the mass in Einstein's equation might be considered to have a single quantum of energy.

The issue of how we define mass is coupled with a debate about how we measure it. Currently we measure mass by comparing an object to the international prototype kilogram. The standard unit of mass, the kilogram, is defined as being equal to the mass of the international prototype kilogram. This prototype is a platinum–iridium alloy but it has been found that its mass varies due to a number of reasons, including collecting dirt from the atmosphere. Other shortcomings with using an object as a standard are that copies need to be made for use around the world and subunits in smaller sizes (such as 500 g, 100 g and 50 g objects) need to be made. This copying and dividing process introduces errors.

The earliest basis for measuring mass was essentially placing an object of known mass in a pan on one side of a balance arm pivoted at its centre and the object to be weighed in a second pan on the opposite side of the pivot arm of the balance. Modern electronic balances are essentially based on this principle.

To overcome these shortcomings, the International Committee on Weights and Measures is considering a plan to redefine the kilogram on the basis of unchanging quantum properties of nature rather than a physical object. The watt balance is an instrument that may allow this to be done.

The watt balance can express a value for mass in terms of a current measured against the force of gravity. It operates on the principle that a conducting wire of known length carrying a current will experience a force when in a perpendicular magnetic field. The current is varied so that the force experienced by the conductor precisely counteracts the weight force of the mass.

Another type of watt balance, the Kibble balance, does away with the need to measure the length of the conductor and the strength of the magnetic field by moving the conducting wire through the magnetic field at a constant known velocity. By Faraday's law of electromagnetic induction, an EMF is induced across the ends of the conductor. Thus, with accurate measurement of the induced EMF, current, velocity of the conductor and the local gravitational field, an accurate mass can be determined.

The advantage of using a watt balance is that current and voltage are defined in terms of universal physical constants such as the speed of light and Planck's constant; thus, this will provide a definition of the kilogram in terms of these constants rather than a physical object.

a Why are physicists trying to find a replacement for the international prototype kilogram? (2 marks)

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b Give Einstein's equation for the relationship between mass and energy. (1 mark)

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c The article states that 'mass … might be considered to have a single quantum of energy'. Derive a mathematical expression to show how mass may be expressed to have a quantum of energy. (2 marks)

d Use a diagram to show the basis for the measurement of mass using the international prototype kilogram as described in the information given. Label the diagram to indicate the appropriate physics concepts for the basis of the measurement for a mass. Explain the principles on which this system operates to find the value of an unknown mass. (5 marks)

e In paragraph six, what does the information state that mass may be expressed in terms of? (1 mark)

f Using a relationship in your study of electromagnetism, derive the mathematical expression for mass in terms of the variables given in paragraph six. (3 marks)

g Using the information given in paragraph seven and the mathematical expression from (f), derive an expression for mass based on current and voltage. (3 marks)

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Additional working space